

SPEAKER PRESENTATION · BSIDES INDORE 2026

Beyond the Internet: Practical Security of LoRa Mesh Networks with Meshtastic

[ABIMEL S B KULUMALA] · [Track 5 : CYBER PHYSICAL, IOT & INDUSTRIAL SECURITY]

45 MIN SLOT · 40 MIN TALK + 5-15 MIN Q&A

ABOUT YOU

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ABSTRACT & KEY TAKEAWAYS

ABSTRACT (AS SUBMITTED AT CFP)

Explore the security of LoRa-based Meshtastic networks through practical experiments covering encryption, metadata exposure, rogue nodes, device compromise, and network resilience. Learn how to identify real-world attack surfaces and apply practical security measures to build more secure off-grid communication networks.

TARGET AUDIENCE

This session is aimed at cybersecurity professionals, penetration testers, security researchers, and IoT/embedded security enthusiasts interested in LoRa and off-grid communication. It is also suitable for students and beginners in wireless security who want practical exposure to Meshtastic and its security challenges.

KEY TAKEAWAYS

Attendees will understand how LoRa mesh networks and Meshtastic handle encryption, routing, metadata, and trust. They will learn to identify common security risks and apply practical techniques for hardening and securely deploying off-grid mesh networks.

WHAT WENT WRONG — THE PIVOT — THE BREAKTHROUGH

01

What Went Wrong

We initially focused on whether Meshtastic's encryption could protect our messages, assuming encryption was the primary security concern. We soon realized that this approach overlooked the larger attack surface around configuration, metadata, device compromise, and network behavior.

02

The Pivot

We shifted from simply testing how the mesh communicates to thinking like an attacker: What can I discover, manipulate, or compromise without breaking the encryption?

03

The Breakthrough

Our experiments showed that securing an off-grid mesh is not just about cryptography; trust, key management, metadata, physical security, and operational security are equally important.

Beyond the Internet

What happens when the Internet, cellular networks, and Wi-Fi disappear?

Why Off-Grid Communication Matters


HELTEC


UNSET
2.5.10
d012
meshtastic.org




HELTEC

What Is a Mesh Network?

A mesh network allows devices to communicate with one another and potentially relay traffic for other devices instead of depending on a single central access point. If Node A cannot directly reach Node D, intermediate nodes can help move the packet through the network.

Example: Node A ⟶ Node B ⟶ Node C ⟶ Node D

This creates resilience because communication does not necessarily depend on one central device. But every additional node also creates another component that must be trusted, configured, and protected.

LoRa vs LoRaWAN vs Meshtastic

These three terms are often used together even though they describe different things.

LoRa is the physical-layer radio technology that enables long-range, low-power communication. LoRaWAN is a networking architecture designed primarily for IoT deployments using gateways and network servers. Meshtastic, on the other hand, is an open-source decentralized mesh communication system that uses LoRa radios to allow nodes to communicate and relay traffic without traditional infrastructure.

What Is Meshtastic?

Meshtastic turns affordable LoRa-capable hardware into nodes that can participate in an off-grid mesh. Nodes can send messages, exchange information, relay packets, and provide features such as position and telemetry while operating without a cellular or Internet connection.

From a security research perspective, this gives us something very interesting: a real communication network that we can build, configure, break, observe, and secure using inexpensive hardware.

Meshtastic Terminology You Need to Know

Node - an individual Meshtastic device participating in the mesh.

Channel - a logical communication group with its own configuration and encryption key.

PSK - the pre-shared key used for channel encryption.

Primary/Secondary Channel - the primary channel is the main channel; additional channels can provide separate communication groups.

Hop Limit - controls how many times a packet can be relayed.

Meshtastic Terminology You Need to Know

Node ID / Node Number — identifiers associated with nodes.

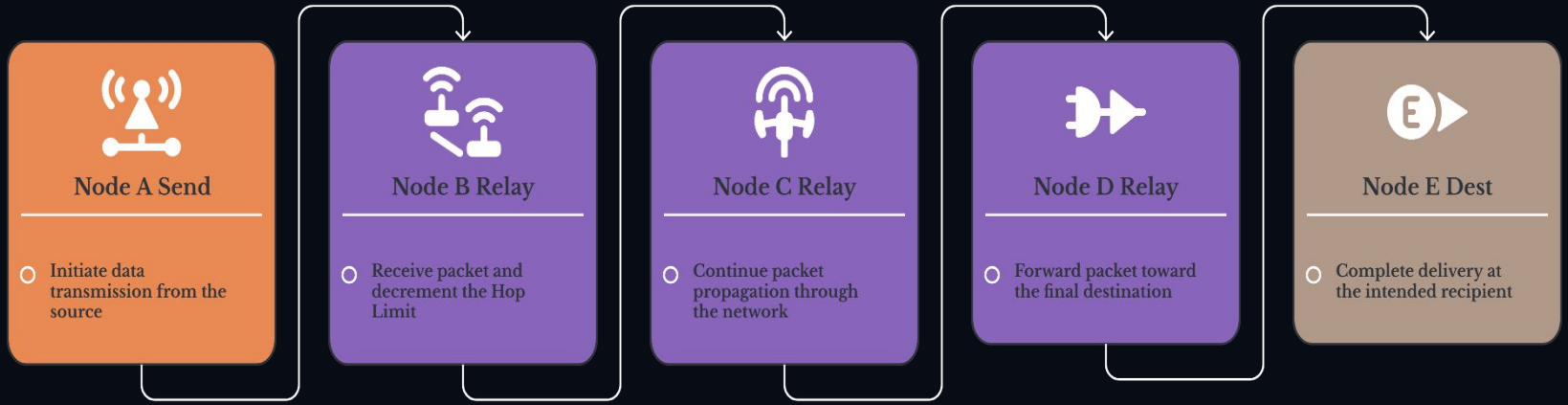
NodeDB — the local database of nodes a device has learned about.

Client / Router / Repeater and other roles — different device behaviors that affect how a node participates in the network.

MQTT — an optional bridge that can connect mesh traffic to Internet infrastructure, which changes the threat model.

How Does a Meshtastic Mesh Actually Work?

LoRa Packet Relay Sequence



Where Is the Security?

At this point we stopped asking, "Does Meshtastic encrypt messages?"

The more useful question became: "What exactly are we protecting?"

We identified several security layers: message confidentiality, key management, node identity, device security, metadata, network availability, routing behavior, physical security, and operational security. Encryption protects an important part of the communication, but it is only one layer of the overall system.

Understanding Meshtastic Encryption

Meshtastic supports channel encryption using a pre-shared key, with AES-128 or AES-256 configurations. All nodes that need to communicate on a channel must possess the appropriate key, making key management a critical part of the security model. Current protocol documentation also describes public-key-based protection for direct messages.

This creates an important security boundary:

If you have the key → potentially decrypt the protected traffic.

If you do not have the key → the encrypted payload should not be readable through normal channel decryption.

The Default Configuration Problem

Meshtastic's protocol includes predefined shorthand PSKs, including a default channel key that is documented in the source code. A deployment that relies on a publicly known default key should therefore not be treated as a private communication system. For sensitive deployments, a properly generated private key and controlled key distribution are essential.

Attack Surface: Think Like an Attacker

Once we understood how the mesh works, we started mapping its attack surface from an attacker's perspective. Instead of trying to break LoRa or the underlying cryptography, we looked at weaknesses in configuration, exposed keys, node identities, location and network metadata, compromised or stolen devices, routing and network availability, poor key handling, user mistakes, OPSEC, and optional bridges to the Internet. This showed us that the most interesting attacks are often not against the cryptography itself, but against the way the network is **configured, deployed, trusted, and operated**.

LoRa Attack Surface

Devices

Lost, stolen, or compromised nodes.

Network behavior

Routing, rebroadcasting, congestion, and availability.

Identity

Node names, IDs, location information, and network relationships.

Users

Poor key handling, screenshots, shared configurations, and OPSEC mistakes.



Configuration

Insecure channels, exposed keys, and unsafe settings.

Bridging

Optional connections between the mesh and Internet infrastructure.

Attack Scenario:

The Unauthorized Node

What If a Node Is Stolen?

Encryption Doesn't Hide Everything

Encrypted communication is not necessarily anonymous communication.

Depending on the configuration and features being used, information such as node identities, communication timing, network relationships, signal information, routing behavior, and location-related data can become valuable metadata.

An attacker may not know what Alice said to Bob, but they may still learn that Alice and Bob are communicating, that several nodes form a particular topology, or that a node is associated with a particular location.

Availability: What Happens When Nodes Disappear?

Defending the Mesh

Use strong, privately generated channel keys instead of relying on publicly known defaults. Protect physical devices, minimize unnecessary identifying information, carefully configure location sharing, separate communication groups where appropriate, maintain an inventory of deployed nodes, keep firmware maintained, and establish a process for responding to compromised devices.

Meshtastic itself recommends carefully choosing node roles and notes that unnecessary hop counts can create network issues, showing that secure deployment also involves understanding network behavior rather than simply enabling encryption.

SUMMARY & CALL TO ACTION

When the Internet disappears, the attack surface doesn't.

Off-grid communication gives us resilience, independence, and new possibilities—but it also introduces a different security model.

https://youtu.be/2J_FnATHHLc

https://youtu.be/ywLzaRSK_us

https://youtu.be/z3AxyRX4e8Y

https://drive.google.com/drive/folders/1Yi6sD7sbyPEN33s1-mRD2xXuEWAKkd6K

QUESTIONS?

whoami

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THANK YOU

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